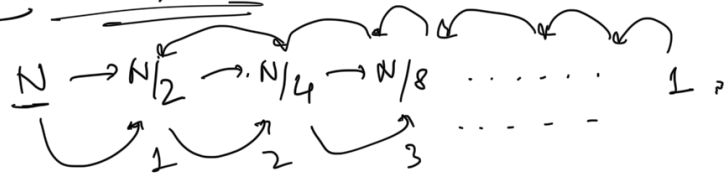


# INTRO TO TIME COMPLEXITY

## MATHEMATICAL PRE-REQUISITES

①



$$\log_2 N$$

$$\log_a N$$

$$N \rightarrow \frac{N}{a} \rightarrow \frac{N}{a^2} \rightarrow \dots$$

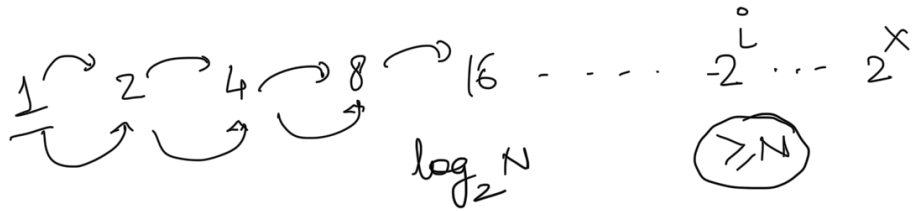
$$\log_2 2^{10} = 10$$

$$\log_{10} 10^5 = 5$$

$$\log_{10} 10^5 = 5 \log_{10} 10$$

$$-\log_b a^x = x \log_b a$$

$$-\log_b a * c = \log_b a + \log_b c$$



$$\log_a N = i$$

$$a^i = N$$

$$N \rightarrow \frac{N}{a} \rightarrow \frac{N}{a^2} \rightarrow \frac{N}{a^3} \dots 1$$

$$\log_2 2^{10} = 10$$

$$2^{10} \rightarrow 2^9 \rightarrow 2^8 \rightarrow 2^7 \rightarrow \dots 2^0$$

10 steps

$$\log a^x = x \log a$$

$$\log ab = \log a + \log b$$

log

$$\log_2 2^{10} = 10$$

$$2^{10} \rightarrow 2^9 \rightarrow 2^8 \rightarrow 2^7 \rightarrow \dots$$

$$2^0 \rightarrow 2^1 \rightarrow 2^2 \rightarrow 2^3 \rightarrow 2^4 \rightarrow \dots$$

$$2^0, 2^1, 2^2, \dots, 2^{10}$$

$$a^i = N$$

$$\log_a N = i$$

$$a^i = N$$

$[a, b]$

$[ ]$

$( )$

↓  
inclusive

↓  
exclusive

$$[3, 5] \rightarrow 3, 4, 5$$

$$(3, 5) \rightarrow 4$$

$$[3, 5) \rightarrow 3, 4$$

$$(2, 5]$$

$$(3, 5] \rightarrow 4, 5$$

$$[3, 5] \rightarrow 3, 4, 5$$

$$[3, 10] \rightarrow 10 - 3 + 1 = 8$$

$$[x, y] \rightarrow (x, x+1, x+2, \dots, y) \rightarrow y - x + 1$$

$$[a, b] \rightarrow b - a + 1$$

$$[a, b] = b - a + 1$$

Arithmetic progression

$$a, a+d, a+2d, a+3d$$

$$2, 3, 4, 5, 6, 7, 8$$

$$1, 3, 5, 7, 9, 11, 13$$

$$1, 2, 3, 4, 5$$

$$\frac{n(n+1)}{2}$$

$$a, a+d, a+2d, a+3d, a+4d, \dots$$

$$\frac{n(n+1)}{2}$$

$$a + (a+d) + (a+2d) + (a+3d) + \dots + (a+(n-1)d)$$

$$= \underline{a \times n} + (\underline{0} + \underline{d} + \underline{2d} + \underline{3d} + \underline{4d} + \underline{5d} + \dots + (n-1)d)$$

$$= a \times n + d(0 + 1 + 2 + 3 + 4 + 5 + \dots + (n-1))$$

$$= a \times n + d \times \frac{n \times (n-1)}{2} = \boxed{\frac{1}{2} \times [2an + dn(n-1)]}$$

$$a \quad a+d$$

$$1 \quad 2$$

$$0 \quad (1) \quad (2) \quad 3$$

$$= \frac{n}{2} [2a + (n-1)d]$$

$$= \frac{n}{2} \left[ \underset{\substack{\uparrow \\ \text{first term}}}{a} + (a + \underset{\substack{\uparrow \\ \text{last term}}}{(n-1)d}) \right]$$

$$= \frac{n}{2} \times [\text{first term} + \text{last term}]$$

Geometric progression

(8)

$r = 1/2$

$$a, ar, ar^2, ar^3, ar^4, \dots, ar^{n-1}$$

$$1, 2, 4, 8, 16, 32, 64, \dots$$

$$1, 1/2, 1/4, 1/8, 1/16, 1/32, 1/64, \dots$$

$$S = \cancel{a} + \cancel{ar} + \cancel{ar^2} + \cancel{ar^3} + \dots + \cancel{ar^{n-1}}$$

$$\downarrow$$

$$rS = ar + ar^2 + ar^3 + \dots + ar^{n-1} + \cancel{ar^n}$$

$$rS - S = ar^n - a$$

$$S(r-1) = a(r^n - 1)$$

$$S = \frac{a(r^n - 1)}{(r-1)} \quad (r \neq 1)$$

$$a \frac{(r^{\text{no. of terms}} - 1)}{r - 1}$$

$$S = 1 + 2 + 4 + 8 + 16 + \dots + 2^{n-1}$$

$$2S = 2 + 4 + 8 + 16 + \dots + 2^{n-1} + 2^n$$

$$S = \frac{2^{n+1} - 2}{2} = 2^n - 1$$

$$1 + 2 + 4 + 8 + \dots + 2^{n-1} = 2^n - 1$$

$$S = N + \frac{N}{2} + \frac{N}{4} + \frac{N}{8} + \frac{N}{16} + \dots + \frac{N}{2^{n-1}}$$

$$\frac{1}{2}S = \frac{N}{2} + \frac{N}{4} + \frac{N}{8} + \frac{N}{16} + \frac{N}{32} + \dots + \frac{N}{2^n}$$

$$S - \frac{1}{2}S = N \Rightarrow \frac{1}{2}S = N$$

$$S \left(1 - \frac{1}{2}\right) = S \cdot \frac{1}{2}$$

$$S = 2N$$

$$\left(1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \dots + \frac{1}{1024}\right) < 2$$

$$\frac{\infty}{2} = \infty$$

3

$$[a, b] = b - a + 1$$

$$(i = 1; i \leq N; i++)$$

$$\begin{matrix} i = 1 \\ i = 2 \\ i = 3 \end{matrix}$$

$$i = [1, N]$$

$$N - 1 + 1 = N$$

fn(N, M):

$$[N + M]$$

$$\begin{aligned} & (i \rightarrow 1, i \leq N, i++) : [1, N] \\ & \text{if } i \% 2 == 0 \\ & \quad \text{print } i \\ & (j = 1; j \leq M; j++) \\ & \quad \text{if } j \% 2 == 0 [1, M] \end{aligned}$$

0 d print j

fn (N) :

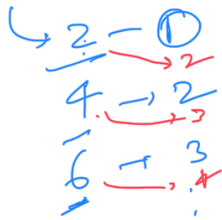
(i = 1; i <= N; i += 2) :

[L, N]

$$\frac{N+1}{2}$$

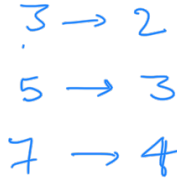
$$\frac{N+1}{2}$$

even



1 3 5 7 9

odd



1, 1+d, 1+2d, ..., N

xth term

1+(x-1)d

i=1  
i=3  
5

1, 1+d, 1+2d, ..., 1+(steps-1)d

$$1 + (steps-1)d = N$$

$$(steps-1)d = N-1$$

$$(steps-1) = \frac{N-1}{d}$$

$$steps = \frac{N-1}{d} + 1$$

$$d=2$$

$$\frac{N-1}{2} + 1$$

$$= \frac{N+1}{2}$$

i=0,

i=1

i=0; i <= 100; i++

[0, 100]

$$100 - 0 + 1 = 101$$



$$\underline{1 + (\frac{N-1}{d})d = N}$$

$$(x-1)d = N-1$$

$$x-1 = \frac{N-1}{d}$$

$$x = \left( \frac{N-1}{d} + 1 \right)$$

$$\frac{N-1}{2} + 1 = \left( \frac{N+1}{2} \right)$$

$$\{ \underline{i=1}; \underline{i*i} \leq N; i++ \}$$

$\underline{\underline{=}}$

$$[1, \sqrt{N}]$$

$$\sqrt{N-1} + 1 = \underline{\sqrt{N}}$$

}

$$i=1$$

$$i=2$$

$$3$$

$$4$$

$$5$$

$$\vdots$$

$$\underline{i}$$

$$6 \leftarrow$$

$$\underline{\underline{i^2 > N}}$$

$$\underline{\underline{6^2 > N}}$$

$$\boxed{\begin{array}{l} i^2 > N \\ \underline{\sqrt{i^2}} > \sqrt{N} \\ i > \underline{\sqrt{N}} \end{array}}$$

$$i=N$$

$$\text{while}(i > 1) \{ i = i/2;$$

\}

\*

$$\begin{array}{l} N \\ N/2 \\ N/4 \\ N/8 \\ \vdots \\ 1 \end{array} \left. \vphantom{\begin{array}{l} N \\ N/2 \\ N/4 \\ N/8 \\ \vdots \\ 1 \end{array}} \right\} \log_2 N$$

$$1, 2, 4, 8, 16, 32, 64 \dots (2^x)$$

$$\underline{i = i * 2}$$

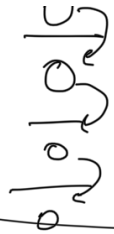
$$\approx \log_2 N$$

$$2^x = N$$

$$x = \log_2 N$$

$$i=0; \underline{i} \leq N; i = i * 2$$

infinity →



$$i \rightarrow [1, 10]$$

$$(j=1; j \leq N; j++)$$

| i   | j      |     |
|-----|--------|-----|
| 1   | [1, N] | N   |
| 2   | [1, N] | N   |
| 3   | [1, N] | N   |
| ... | ...    | ... |

$$10 \quad N-1-0+1 \quad N \quad 10N$$

$$(i=0; i < N; i++) \{ [0, N-1]$$

$$(j=0; j < N; j++) \{$$

|       |     |
|-------|-----|
| i=0   | N   |
| i=1   | N   |
| i=2   | N   |
| ...   | ... |
| i=N-1 | N   |

$$N+N+N+\dots$$

$$N^2$$

$$(i=1; i \leq N; i++) \{$$

$$(j=1; j \leq N; j=j*2) \{$$

$$= \text{sum}++;$$

| i   | j              | log i |
|-----|----------------|-------|
| i=1 | 1, 2, 4, 8, 16 | log 1 |
| i=2 | ...            | log 2 |
| i=3 | ...            | log 3 |
| ... | ...            | ...   |
| N   | ...            | N log |

$$2^x = N$$

$$x = \log_2 N$$

$$\log N + \log N + \dots + N \log$$

$$= N \log N$$

$$(i=1; i \leq 2^N; i++) \{$$

$$[1, 2^N]$$

$$2^N - 1 + 1 + \dots + 2^N$$

sum = 0

( i = 1; i <= N; i++ ) {

( j = 1; j <= 2<sup>i</sup>; j++ ) {

sum++;

}

}

$N \times 2^N$

| i   | j                    | # of times     |
|-----|----------------------|----------------|
| 1   | [1, 2 <sup>1</sup> ] | 2 <sup>1</sup> |
| 2   | [1, 2 <sup>2</sup> ] | 2 <sup>2</sup> |
| 3   | [1, 2 <sup>3</sup> ] | 2 <sup>3</sup> |
| ... | ...                  | ...            |
| N   | [1, 2 <sup>N</sup> ] | 2 <sup>N</sup> |

Geometric Progression

$$2^1 + 2^2 + 2^3 + 2^4 + \dots + 2^N$$

$$S = \frac{a(r^n - 1)}{(r - 1)} = \frac{2(2^N - 1)}{(2 - 1) = 1} = 2(2^N - 1)$$

$$2(2^N - 1) = \frac{2^{N+1} - 2}{1}$$

i = 1 to N :  
 { j = 1; j <= 2<sup>i</sup>; j++ }  
sum++;  
 }  
 sum

i = 1, j = [1, 2<sup>1</sup>], 2<sup>1</sup>  
 i = 2, j = [1, 2<sup>2</sup>], 2<sup>1</sup> + 2<sup>2</sup>  
 i = 3, j = [1, 2<sup>3</sup>], 2<sup>1</sup> + 2<sup>2</sup> + 2<sup>3</sup>



$$2^1 + 2^2 + 2^3 + 2^4 + \dots + 2^N$$

sum = 0

(i = 1, i <= N, i++) {

(j = 1, j <= i, j++) {

sum++

}

| i | j      | sum                        |
|---|--------|----------------------------|
| 1 | [1, 1] | 1                          |
| 2 | [1, 2] | <u>1 + 2</u>               |
| 3 | [1, 3] | <u>1 + 2 + 3</u>           |
| 4 | [1, 4] | <u>1 + 2 + 3 + 4</u>       |
| N |        | <u>1 + 2 + 3 + 4 + ...</u> |
|   |        | $= \frac{N(N+1)}{2}$       |

(k = 1, k <= 1, k++) {  
= 1 times

$$= \frac{N^2 + N}{2}$$

N  
N/2  
N/4  
N/8  
...  
2

← (i = N, i > 1, i = i/2) {  
(j = 1, j <= i, j++) {  
sum++

8 → 1...8  
4 → 1...4

| i   | j         | sum          |
|-----|-----------|--------------|
| N   | [1...N]   | <u>N</u>     |
| N/2 | [1...N/2] | <u>+ N/2</u> |
| N/4 | [1...N/4] | <u>+ N/4</u> |

$$\begin{array}{l} 2 \rightarrow 1 \dots 2 \quad | \quad +2 \\ 1 \rightarrow 1 \quad \quad \quad | \quad +1 \end{array}$$

$$\begin{array}{c} N/8 \\ \vdots \\ \underline{1} \end{array}$$

$$\begin{array}{c} + N/8 \\ + N/16 \\ \vdots \\ \underline{1} \end{array}$$

$$\text{Sum} = \left( N + \frac{N}{2} + \frac{N}{4} + \frac{N}{8} + \frac{N}{16} + \dots + 1 \right)$$

$$= N \left( 1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \frac{1}{32} + \dots + 1 \right)$$

$$< \underline{\underline{N \left( 1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \frac{1}{32} + \dots \right)}} = 2$$

$$< 2N$$

$$S = 1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \dots \infty$$

$$\text{Sum} = \underline{\underline{N + \frac{N}{2} + \frac{N}{4} + \frac{N}{8} + \dots + \textcircled{1} \frac{N}{2^x}}}$$

$$= N \left( 1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots + \frac{1}{2^x} \right)$$

$$\leq \textcircled{2N}$$

$$S = \textcircled{1} + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \dots \infty$$

$$\frac{1}{2}S = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots \infty$$

$$\frac{1}{2}S = 1, \quad \textcircled{S = 2}$$

$$\boxed{S = 1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \dots \infty}$$

$$= \textcircled{2}$$

no. of iterations =  ~~$4N^2$~~  +  $3N$  +  $2$

$O(N^2)$

$N + 3 \log N$

$O(N)$

~~$1/2$~~

$\frac{N(N+1)}{2} = \frac{N^2}{2} + \frac{N}{2}$

$O(N^2)$

$3 \sqrt{N\sqrt{N}} + 4 \log N + 31 \frac{N \log N}{2}$

$N = 2^{64}$

$2^{64} \times \sqrt{2^{64}}$   
 $\downarrow$   
 $2^{32}$   
 $= 2^{96}$

$O(N\sqrt{N})$

$2^{64} \times \log 2^{64}$   
 $64 \times 2^{64}$

$2^{32} \times 2^{64}$

$N \rightarrow$  all factors of  $N$

$N = a \times b$   
 $a \leq \sqrt{N}$

$\{ i = 1; i \leq \sqrt{N}; i++ \}$

if  $(N \% i == 0)$

$i \rightarrow$  one factor

$N/i \rightarrow$  another factor

}

}

$N/i \neq i$